

Three convergences of Q K (zeros), N (basis), L (window)

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Abstract

Q converges in three parameters, already measured, with three different meanings. In the number K of zeros kept, $Q_K \rightarrow Q^{\text{pr}}$ in Frobenius, with a drop at the band edge and an algebraic tail $O(1/\omega_{\max})$. In the basis size N , $\lambda_{\min}(Q_N) \downarrow c_L > 0$ under RH (the floor). In the window L , $Q_L \succeq c_L > 0$ for all L is RH. Nothing new is computed here; the rates are named.

1 K : the explicit formula as a matrix limit

On V_N at fixed μ ,

$$Q^{\text{pr}} = P + A - \sum_p T_p, \quad Q_K = \sum_{k \leq K} 2 \hat{\eta}(\gamma_k) \hat{\eta}(\gamma_k)^\top.$$

Measured at $\mu = 11$, $N = 47$, $\omega_{\max} = 120.53$:

$$\frac{\|Q^{\text{pr}} - Q_K\|_F}{\|Q^{\text{pr}}\|_F} = \begin{cases} 31\% & K = 35, \gamma = 112, \\ 8.9\% & K = 40, \gamma = 123, \\ 5.1\% & K = 60, \gamma = 163. \end{cases}$$

The drop sits on the band edge. The remainder is the off-band lobes $\hat{\eta}_n(\gamma) \sim 1/(\gamma^2 - \omega_n^2)$.

Lemma 1 (tail). $\sum_{\gamma > \omega_{\max}} |\hat{\eta}_n(\gamma) \hat{\eta}_m(\gamma)| \leq C_N / \omega_{\max}$.

An Arb enclosure of $Q^{\text{pr}} - Q^z$ on the whole V_{47} puts every one of the 1128 independent entries in a ball about 0 of radius 6×10^{-3} (§65). That is convergence in K certified, not a density-1 residual.

2 N : the floor

At fixed L , $V_N \uparrow PW_{L/2}^{\text{even}}$. Under RH, Q is the Gram of the evaluators on that space, so

$$\lambda_{\min}(Q_N) \downarrow c_L = \inf_{\|f\|_2=1} \sum_{\gamma} |\hat{f}(\gamma)|^2 > 0$$

(Beurling / Duffin–Schaeffer on a set of density $< L/\pi$, *sampling-floor*). Measured: at $\mu = 3$, $c_{\log 3} = 5.55 \times 10^{-8}$, ratios $1.32 \rightarrow 1.02$; at $\mu = 11$, $\lambda_{\min}(N) = 3.59, 1.86, 1.54, 1.4 \times 10^{-48}$ for $N = 47, 57, 67, 80$ — flattening, not a new decade per ten modes. The statement “one keeps digging” (§54) is false. Q_N converges as an operator on each compact of $PW_{L/2}$ to the Beurling form; only the razor is numerically ugly.

3 L : the one that is RH

Q_L on $PW_{L/2}$, $L \rightarrow \infty$, asks for a uniform $c_L > 0$. Under RH this is the sampling floor with more and more points; the constant is expected to decay (geometric law, desert). Unconditionally it is Weil positivity at every compact window, i.e. RH. There is no measured limit $Q_L \rightarrow Q_\infty$ in the repository that would bypass that.

Status

K converges, with a named tail and an enclosure. N converges, to a positive floor under RH, measured at two windows. L does not converge in the deposit; that limit is the theorem. Not a new factor of Q . Not RH.